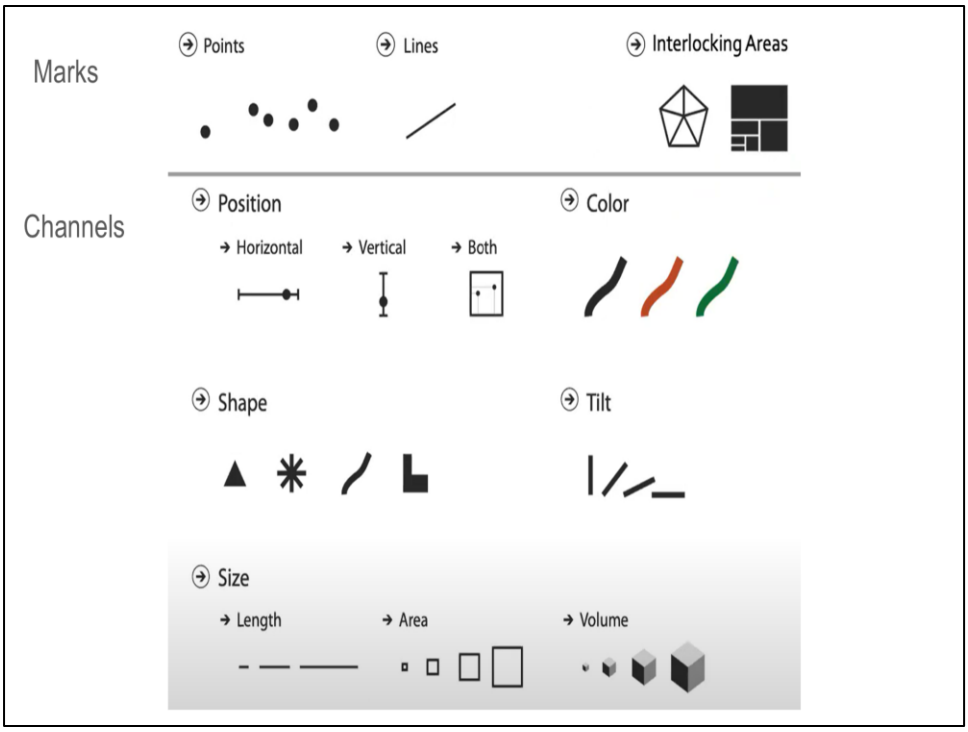
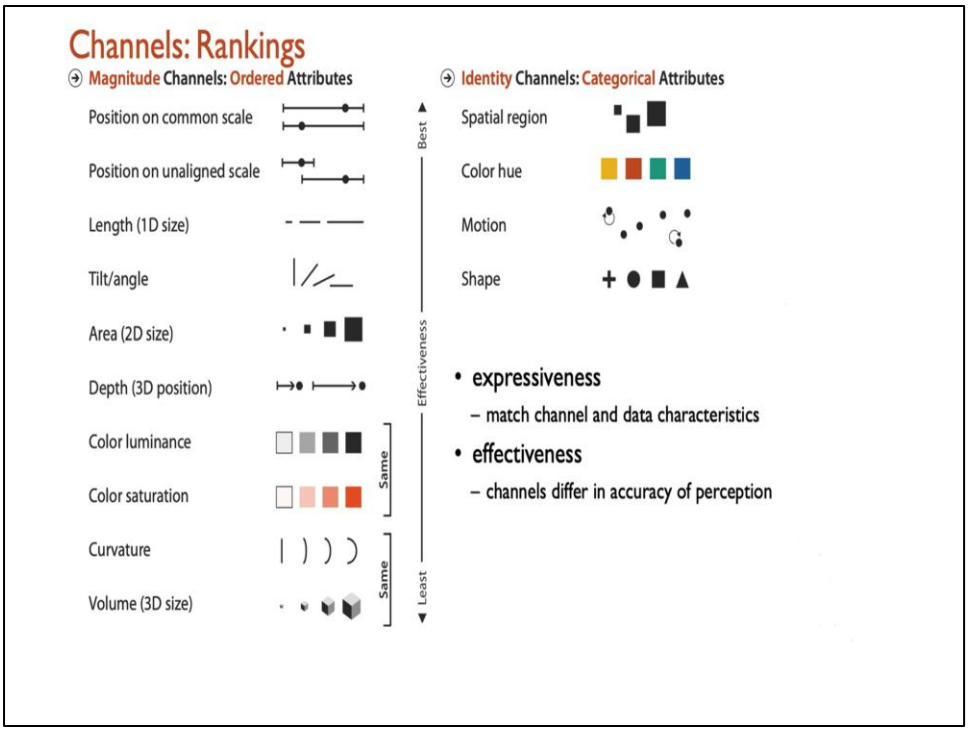
Definitions Up Front

- **marks** represent items or links in a visualization
  - points, lines, areas
  - containment, connection
- **channels** are way to change the appearance of marks based on mapping attribute values
  - type and amount of info represented by channels can differ

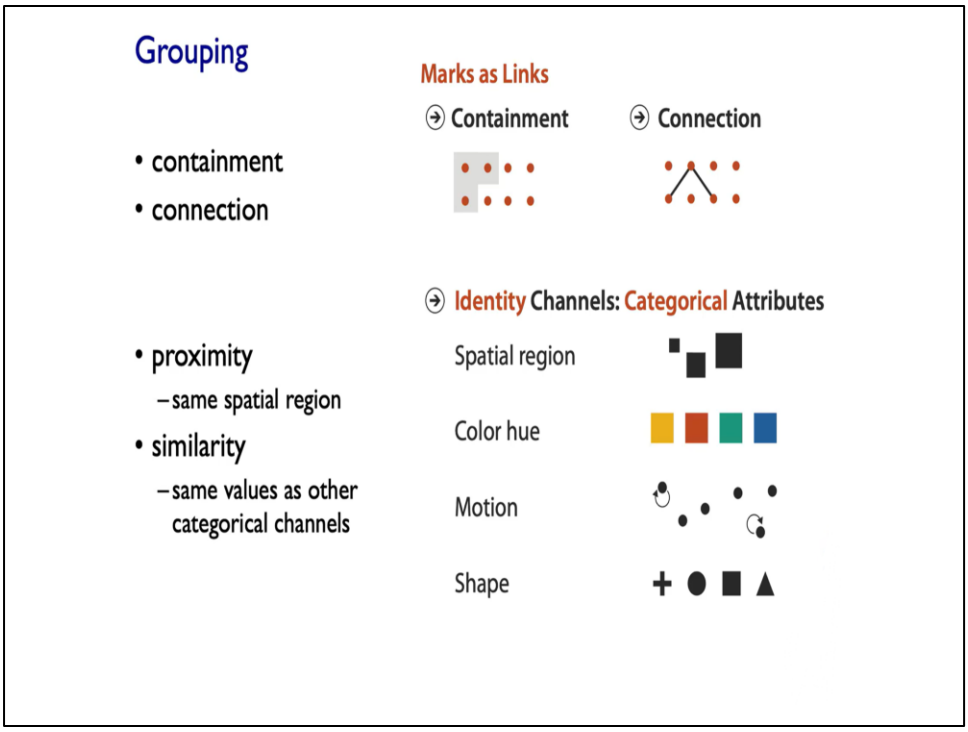
- Marks : Data points of the graph
  - Containment : Similar to a tree map
    - Lines in a line graph
    - Points in a scatter plot
    - Bars in a bar graph
    - Display hierarchy with rectangular areas
  - Connection : Similar to a graph or a network
    - Nodes and line connecting the nodes
- Channels
  - A way to map data elements into visual elements in dataset
    - EX: Different color to identify different groups in graph



- Marks : data points and connections
- Channels : Mapping technique to identify one set from another



- Channel ranking
  - Magnitude channels: Numerical attributes
    - Order attributes
  - Identity rank
    - Categorical attributes
  - Effectiveness : How effective the visual captures the eye
    - Accuracy in displaying information
    - Ultimate goal : how much information the attribute can convey
  - Expressiveness: visualization limited to what the data contains
    - How much of the data does the visualization convey



- Grouping
  - Containment : Area of data in a bin (bar graph)
  - Connection: connection of nodes